Your VLA is Secretly kNN

A nearest-neighbour lookup over the demonstrations, with no training, is a probe for vision-language-action models: it shows how much of a benchmark is retrieval, how good a VLA's backbone is, and what its action expert adds. LIBERO and LIBERO-plus, five LeRobot VLAs.

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91.7%
LIBERO success of a kNN lookup over the demos: no training, on par with a diffusion transformer.

88–94%
of every VLA's successes are also a plain ImageNet lookup's, episode by episode.

−5.3 vs +12.5
what the action expert adds over a lookup on its own VLM: SmolVLA vs $\pi_{0.5}$.

$\rho = 0.90$
an offline neighbour metric ranks backbones by their closed-loop lookup success.

1 A kNN probe

The claim

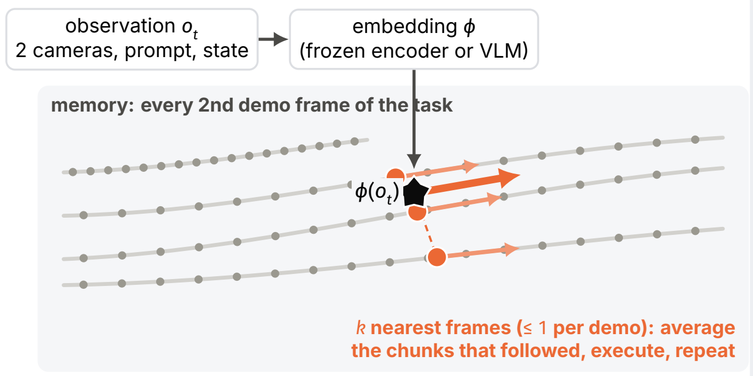
A VLA maps camera images, an instruction and the robot state to a chunk of future actions. We ask whether it effectively factorises into an **embedding** ϕ, its VLM (the *backbone*), and a **lookup** over the demonstration frames $(o_i, a_{t_i:t_i+H})$ it was trained on:

$$\pi(o) \approx \sum_{i \in N_k(\phi(o))} w_i \, a_{t_i:t_i+H}, \qquad w_i \propto e^{-\|\phi(o)-\phi(o_i)\|^2/\tau}$$

A lookup needs no training, so it is a measuring stick for three questions:

1. **How hard is the task?** What a **kNN** over generic image features already solves (§2).
2. **How good is the backbone?** The lookup keyed on the VLA's own VLM, or offline, whether close embeddings have close actions (§3).
3. **How good is the action expert?** What the VLA adds beyond that lookup (§3).

A kNN policy in LeRobot

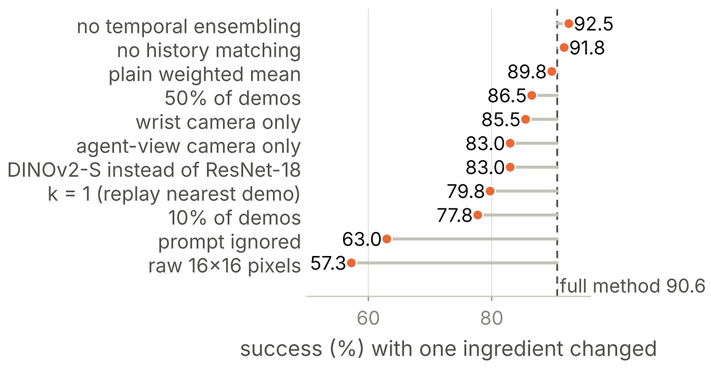


- **Memory**: every 2nd frame of the 1,693 demos (137k keys) with the chunk that followed. "Training" is embedding them: minutes, no gradient step.
- **Key** ϕ: a frozen ImageNet ResNet-18 (layer 3, 4×4 grid, both cameras), or the hidden states of any VLA's VLM (§3).
- **Act**: average the chunks of the k nearest frames of the task, ≤1 per demo (0.5 ms). The *full method* adds state and prompt distances, mode seeking and temporal ensembling; the *plain lookup* none of these.

Setup

LIBERO: 4 suites × 10 tasks, ~42 demos per task, 2 cameras. Tuned on initial states 40–49 and 0–19; every comparison uses the same 400 episodes (20–29); the headline is the official `lerobot-eval` (0–49). **VLAs**: LeRobot Hub LIBERO fine-tunes of SmolVLA (0.45B), X-VLA (0.9B), π_0, π_0-FAST, $\pi_{0.5}$ (3B); a multi-task DiT trained here.

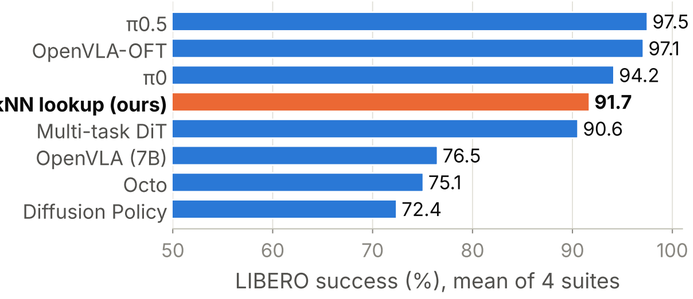
What the lookup needs



The representation, the prompt and the data do the work. Without the prompt, success falls to 63.0% (Goal: 8%, where only the instruction differs), with raw pixels to 57.3%, with a single neighbour to 79.8%, with 10% of the demos to 77.8%; the extras move it by about a point each: **the memory is the model**.

2 How hard is the task?

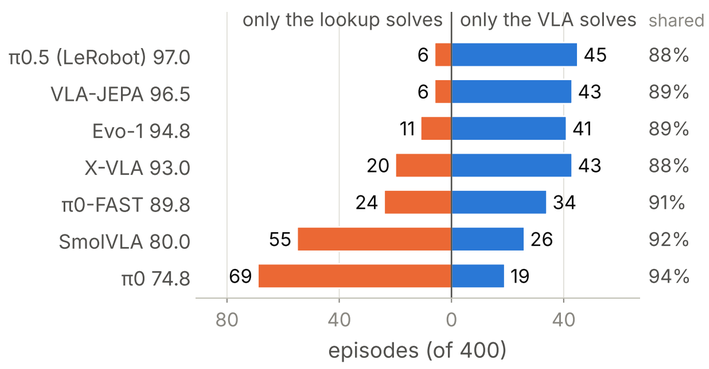
LIBERO rewards retrieval



A lookup with no training is above Diffusion Policy, Octo and OpenVLA and **on par with LeRobot's multi-task diffusion transformer** (90.6%; per suite 88.0/99.8/97.6/81.4). **31 of the 40 tasks** are solved in at least 90% of episodes by the plain lookup over ImageNet features (35 by the full method); the hard ones are long-horizon.

Which episodes need more than a lookup?

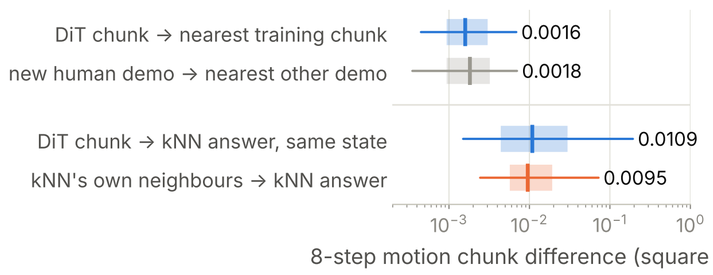
Same 400 episodes, each VLA against the plain ImageNet lookup (87.3%); VLA success after its name.



88–94% of every VLA's successes are also the lookup's. VLAs differ in the rest: where the lookup fails, $\pi_{0.5}$ still succeeds in 88% of the episodes and X-VLA in 84%, SmolVLA in 51% and π_0 in 37%, and the weaker ones also lose episodes the lookup solves (SmolVLA 55, π_0 69).

A diffusion head behaves like kNN

We trained LeRobot's multi-task DiT on the same demos (reduced budget, 68.8% success) and recorded all 789 chunks it generated from unseen initial states.



Its outputs are not new: a DiT chunk is as close to some training chunk (median 0.0016) as a new human demonstration is to the other demos (0.0018). **Its outputs are the kNN answer**: at the same state it is as far from the kNN average (0.0109) as kNN's own neighbours are (0.0095), and the grippers agree 83% of the time.

Off the demonstrations: LIBERO-plus

The same 40 tasks under seven perturbations: every policy runs the same 420 variants (60 per perturbation; not the published leaderboard's set), next to its clean LIBERO success.

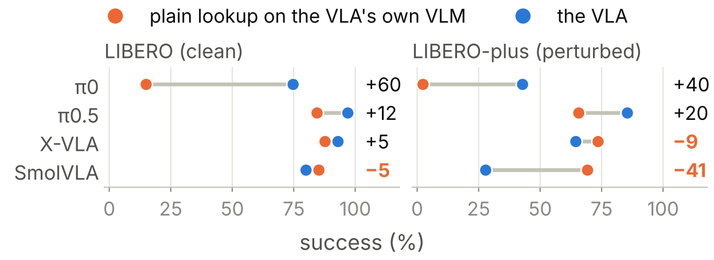
	LIBERO	Camera	Robot	Language	Light	Background	Noise	Layout	all
kNN, full method	91	73	58	98	92	97	92	67	**82**
kNN, plain (ResNet-18)	87	68	48	77	92	85	88	70	**75**
π0.5 (LeRobot)	97	68	78	92	100	93	83	83	**85**
X-VLA	93	7	83	77	93	93	42	57	**65**
π0-FAST	90	10	15	33	43	37	0	42	**26**
SmolVLA	80	25	13	22	43	45	10	37	**28**
π0	75	67	27	12	43	58	52	42	**43**
VLA-JEPA	96	55	58	88	92	93	55	77	**74**

A lookup degrades no more than the best VLA (points lost: kNN 8.2, $\pi_{0.5}$ 11.5, X-VLA 28.5, SmolVLA 52.1, π_0-FAST 64.1). **Weak VLAs key on the wording**: reworded instructions over the original scenes drop SmolVLA, π_0-FAST and π_0 to 22, 33 and 12% (original wording: 82, 88, 68%). X-VLA fails under a camera shift (7%).

3 Backbone vs action expert

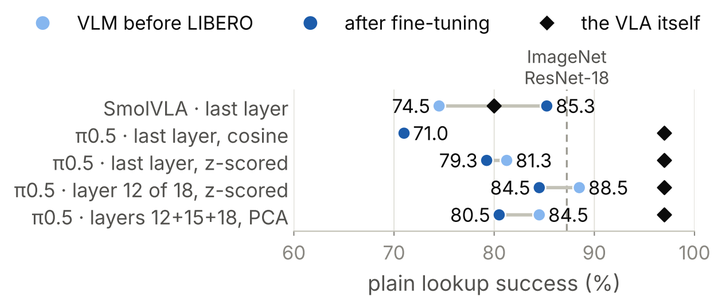
A VLA against a lookup on its own VLM

The plain lookup keyed on a VLA's own VLM scores its backbone; the VLA's excess over it is its action expert's.



SmolVLA's action expert makes it worse than a lookup on its own VLM (80.0% vs 85.3%); $\pi_{0.5}$**'s adds +12.5 points**. Under LIBERO-plus perturbations the gaps widen: SmolVLA −**41**, X-VLA −9, $\pi_{0.5}$ +20. π_0's fine-tune broke its own key (its base VLM keys a 86.0% lookup).

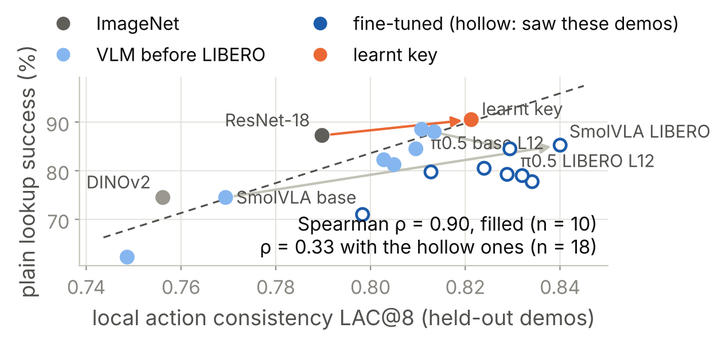
Reading a VLM as a key



A fair backbone score needs the right key. $\pi_{0.5}$ puts ~80% of its energy in 8 of 2,048 channels: standardise every dimension (71.0 → 79.3%) and read layer 12 of 18 (84.5%); a whitened stack of layers 12, 15 and 18 does not help (80.5%). Fine-tuning improves SmolVLA's key, not $\pi_{0.5}$'s (88.5 → 84.5%).

Scoring a backbone without rollouts

A lookup only uses the nearest neighbours, so we score a key by its **local action consistency**: LAC@8 = 1 − action distance to its 8 nearest frames of other demos / to a random frame of the task (held-out queries).



LAC ranks backbones ($\rho = 0.90$ over the 10 keys that never saw these demos; over all pairs, $\rho = 0.31$). Fine-tuned VLMs score higher on their own training demos (hollow). **A key trained for LAC** (one linear map per ResNet grid cell, soft-kNN loss, 90% of demos) lifts it 0.79 → 0.82 and the lookup 87.3 → **90.5%** (full method 91.5%).

Takeaways

- **LIBERO is mostly retrieval**: a lookup with no training reaches 91.7%, and 88–94% of every VLA's successes are also a plain ImageNet lookup's. Under LIBERO-plus perturbations it degrades no more than $\pi_{0.5}$, while the weaker VLAs lose 30–64 points.
- **Small VLAs are secretly kNN**: SmolVLA's action expert is worse than a lookup on its own VLM (−5.3); $\pi_{0.5}$'s adds +12.5, most on long tasks.
- **Backbones can be scored offline**: local action consistency ranks them ($\rho = 0.90$), and a key trained for it lifts the lookup 87.3 → 90.5%.
- Report a **kNN baseline** with every VLA result on a dense-demonstration benchmark.

Limits. LIBERO has about 42 demos per task and test scenes close to them, so it rewards retrieval. Nothing here says a lookup generalises to new objects, scenes or tasks; that is exactly what a good ϕ has to provide.

Code: `lerobot_policy_knn`, a LeRobot policy plugin (`uv sync --extra libero --group knn`); every number and figure is regenerated from the experiment logs by `make`.

Baselines: OpenVLA, OpenVLA-OFT papers, LeRobot docs. VLAs: LeRobot Hub checkpoints.